

MODULE 37 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Database Migration

Schema and data are two migrations. The effort hides in the first.

DOMAIN 4 of 4 EXAM WEIGHT 20% MODULE 37 of 44

Moving the rows is the solved problem.
**Converting the logic is where migrations
overrun.**

Homogeneous or heterogeneous?

HOMOGENEOUS

Same engine, both ends

- Oracle to Oracle, MySQL to MySQL
- No schema conversion needed at all
- DMS runs it serverless, using native tools
- Carries functions, triggers and partitions

HETEROGENEOUS

The engine changes

- Oracle to Aurora PostgreSQL, and similar
- Schema must be converted first
- Code objects need review and rewriting
- The data move is the easy half

THE TIEBREAKER

Homogeneous migrations need no conversion step at all, which is why they are dramatically faster to plan and to execute.

The three task modes

- ◆ **Full load** copies existing data once. Suitable when the source can be frozen for the duration.
- ◆ **Full load plus CDC** copies everything, then keeps applying changes. This is the minimal-downtime pattern.
- ◆ **CDC only** applies ongoing changes to a target you populated some other way, such as a native backup restore.
- ◆ You cut over when replication lag is near zero, not when the full load finishes. Those are different moments.

ON THE EXAM

Change data capture is what turns a maintenance weekend into a few minutes. Without it, downtime equals the whole transfer.

Converting the schema

- ◆ **AWS SCT** is the standalone client you install locally, and it has **more features**.
- ◆ **DMS Schema Conversion** is built into the DMS console, and has **fewer features** than the client.
- ◆ AWS's own wording: it converts "**a majority of the database code objects**". A majority is not all of them.
- ◆ The assessment report lists what could not be converted, with an effort estimate. Read it before committing to a date.

ON THE EXAM

Read as "everything", the remaining work looks like review. Read correctly, it is rewriting, and that changes the plan.

Proving it arrived

- ◆ **Premigration assessment** checks the source and target for problems before you start, such as unsupported data types.
- ◆ **Data validation** compares source and target rows during and after migration, reporting mismatches per table.
- ◆ A completed task is not a validated task. Completion means DMS finished; validation means the data agrees.
- ◆ Keep the source running and replicating after cutover, so reversing is possible if something is wrong.

ON THE EXAM

Validation is the answer whenever a scenario says row counts differ or data looks wrong. Do not reach for a manual reconciliation script.

Two honest notes

- ◆ **DMS Fleet Advisor support ended on 20 May 2026.** It stopped taking new customers a year earlier.
- ◆ **AWS names Migration Evaluator as its successor** for database portfolio assessment, as in module 33.
- ◆ **DMS itself is unaffected and fully supported.** Fleet Advisor was one capability, not the service.
- ◆ **DMS Serverless** removes replication instance sizing for heterogeneous work, scaling capacity automatically.

ON THE EXAM

Fleet Advisor answered which databases to migrate and how hard each would be. Migration Evaluator now answers that.

Reading the scenario

THE SCENARIO SAYS

Oracle to Aurora PostgreSQL

Oracle to RDS for Oracle

Minimal downtime at cutover

Target already populated from a backup

Row counts do not match after migration

Which databases should we migrate, and how hard?

IT IS POINTING AT

Heterogeneous: SCT converts, DMS moves

Homogeneous: no conversion needed

Full load plus CDC

CDC only

DMS data validation

Migration Evaluator, not Fleet Advisor

ON THE EXAM

Ask whether the engine changes first. That single fact decides whether there is a conversion project at all.

Meridian Freight must move a 4 TB Oracle database to Aurora PostgreSQL. The application is being rewritten separately. Downtime must not exceed 30 minutes, and the database serves continuous freight bookings.

The team has scheduled a full load over a weekend.

What should the solutions architect recommend?

- ☐ A Convert the schema with SCT, then run a DMS full load plus CDC task, cutting over when replication lag is near zero
- ☐ B Run the full load over the weekend as planned
- ☐ C Use DMS CDC-only, since the target is empty
- ☐ D Export with Oracle Data Pump and import into Aurora PostgreSQL

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** SCT, then full load plus CDC
The engine changes, so conversion comes first. Full load plus CDC keeps the target current, so cutover is minutes rather than the transfer time.
- B** Weekend full load
Downtime equals the whole transfer, and bookings are continuous. A weekend is far more than 30 minutes.
- C** CDC only
CDC-only applies ongoing changes to a target populated another way. An empty target has nothing for changes to apply to.
- D** Data Pump export and import
Oracle's native tools produce Oracle format. They cannot load into PostgreSQL, and there is no ongoing replication.

THE SIGNAL

Two requirements: the engine changes, and downtime is capped. The first needs SCT and the second needs CDC.

A company is moving a self-managed PostgreSQL database on EC2 to Amazon RDS for PostgreSQL. It contains 200 stored procedures, 40 triggers and several partitioned tables. A junior architect has scoped two months for schema conversion using SCT.

The database is 800 GB.

What should the solutions architect advise?

- ☐ A This is homogeneous, so no schema conversion is required; use DMS homogeneous migration, which carries secondary objects
- ☐ B Proceed with the two-month SCT conversion as scoped
- ☐ C Rewrite the stored procedures for RDS compatibility
- ☐ D Use full load plus CDC with SCT running in parallel

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A** Homogeneous; no conversion needed
PostgreSQL to PostgreSQL means the dialect is identical. DMS homogeneous migrations are serverless and carry procedures, triggers and partitions.
- B** Two-month conversion
Two months of effort converting PostgreSQL into PostgreSQL. The schema already runs on the target engine unchanged.
- C** Rewrite the procedures
They are already PostgreSQL procedures. RDS runs the same engine, so there is nothing to make compatible.
- D** CDC with SCT in parallel
Correct on the data movement, and still budgets conversion work that this migration does not require.

THE SIGNAL

Same engine at both ends is the whole answer. Self-managed to managed is a hosting change, not an engine change.

An SCT assessment report for an Oracle to Aurora PostgreSQL migration shows that 78% of code objects converted automatically. The project plan allocates two weeks for "reviewing the converted schema" and assumes the remaining objects need minor adjustment. There are 1,400 code objects in total.

What should the solutions architect advise?

- ☐ A Roughly 300 objects need manual rewriting, not review; re-estimate the plan from the assessment report's effort ratings
- ☐ B Accept the plan; 78% is a good conversion rate
- ☐ C Re-run SCT with different settings to raise the conversion rate above 95%
- ☐ D Migrate the unconverted objects with DMS instead

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A** About 300 need rewriting; re-estimate
22% of 1,400 is roughly 300 objects requiring human work. The assessment report rates each by effort, which is what a plan should be built on.
- B** 78% is good
It is a good rate, and the plan misreads the other 22% as adjustment rather than rewriting. Good rate, wrong estimate.
- C** Re-run with new settings
Settings do not make an Oracle package with no PostgreSQL equivalent convertible. The gap is dialect, not configuration.
- D** Move them with DMS
DMS migrates data, not code objects. It has no mechanism for converting a stored procedure.

THE SIGNAL

AWS says "a majority" of code objects convert. This scenario is what that phrase means once you multiply it out.

A DMS task reports as complete. Spot checks by an analyst find that three tables have fewer rows in the target than the source, and one has NULL values where the source had data. The team is preparing a manual reconciliation script.

The migration is Oracle to Aurora PostgreSQL.

What should the solutions architect recommend?

- ☐ A Enable DMS data validation, which compares source and target rows and reports mismatches per table
- ☐ B Write the reconciliation script as planned
- ☐ C Re-run the full load from the beginning
- ☐ D Accept the differences, since DMS reported the task complete

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** DMS data validation
Built for exactly this: it compares rows between source and target and reports mismatches per table, continuously, without anyone writing code.
- B** Reconciliation script
Rebuilds a feature DMS already has, on a deadline, and a hand-written comparison across a large schema is itself a source of errors.
- C** Re-run the full load
Repeats the migration without knowing what went wrong, so the same unsupported data types or missing keys produce the same result.
- D** Accept it
Complete means DMS finished. It does not mean the data agrees, which is precisely the distinction being tested.

THE SIGNAL

A completed task and a validated task are different claims. Complete is about the tool; validated is about the data.

A 20 TB SQL Server database must move to RDS for SQL Server. Restoring a native backup to the target would take about four hours, which the business accepts. However, the source will continue receiving writes for two weeks afterwards while an application is tested. The two databases must stay in step during that fortnight.

What should the solutions architect recommend?

- A Restore the native backup, then run a DMS CDC-only task to apply ongoing changes for the fortnight
- B A DMS full load plus CDC task instead of the backup restore
- C A DMS full load task, repeated nightly for two weeks
- D Restore the backup and re-restore it at the end of the fortnight

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Native restore, then CDC-only
CDC-only exists for exactly this: a target populated by other means, kept in step by DMS thereafter. It is the fastest path for 20 TB.
- B** Full load plus CDC
Would work, and DMS full load of 20 TB is slower than a native restore the business has already accepted.
- C** Nightly full loads
Reloads 20 TB every night for a fortnight, and the target is stale for most of each day between runs.
- D** Restore twice
Leaves the target two weeks out of date throughout the testing, which is the requirement it fails.

THE SIGNAL

This is the scenario CDC-only exists for, and it appears rarely. Native tooling loads faster; DMS keeps it current.

A company must assess 240 databases across its estate to decide which to migrate, what each would cost on AWS, and how complex each conversion would be. An architect has proposed AWS DMS Fleet Advisor, having used it on a previous project.

The project starts next month.

What should the solutions architect advise?

- ☐ A Fleet Advisor support ended in May 2026; use Migration Evaluator for the portfolio assessment
- ☐ B Proceed with Fleet Advisor as proposed
- ☐ C Use DMS Schema Conversion assessment reports for all 240 databases
- ☐ D Assess manually with a questionnaire to each database owner

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

Fleet Advisor has ended; use Migration Evaluator

Support ended 20 May 2026, and AWS names Migration Evaluator as the successor. It also produces the cost comparison the scenario asks for.
- B

Use Fleet Advisor

The console and resources are no longer accessible. It stopped accepting new customers a year before support ended.
- C

Schema Conversion reports

Gives conversion complexity per database, and answers neither which to migrate nor what each would cost.
- D

Questionnaires

Asks owners to estimate conversion complexity and AWS cost, which is what the tooling exists to calculate.

THE SIGNAL

DMS is fine. Fleet Advisor was one capability within it, and only that capability has gone.

What to carry out of this module

- ◆ **Ask whether the engine changes.** Homogeneous needs no conversion at all, which changes the timeline completely.
- ◆ **Schema conversion handles "a majority" of code objects.** Multiply the remainder out before agreeing a date.
- ◆ **Full load plus CDC is the minimal-downtime pattern.** CDC-only is for a target populated some other way.
- ◆ **Complete is not validated.** Data validation compares the rows; task completion only means DMS finished.

ON THE EXAM

Fleet Advisor's support ended in May 2026, and Migration Evaluator replaces it. DMS itself is unaffected.